

EDA Using Power BI

Project Title: Marketing Campaign Effectiveness

Problem Statement :

The marketing team runs campaigns across email, social media, TV, and Search Ads. Although marketing spend is increasing, management is unsure whether campaigns are delivering a good return on investment (ROI).

The leadership team wants to know:

- Which channels generate the highest revenue
- Which campaigns underperform
- How efficiently marketing money is spent

You are responsible for building a campaign performance analytics solution to support smarter marketing investments.

Questions:

1. Identify missing or inconsistent marketing data.
2. Which marketing channels generate the highest revenue?
3. Calculate ROI for each campaign using DAX.
4. Analyse conversion rates across channels.
5. Perform univariate analysis on campaign spend.
6. Identify underperforming campaigns.
7. Design a campaign performance dashboard.
8. Which channels should receive increased investment?
9. How can insights improve future marketing strategy?

1) Identify missing or inconsistent marketing data ?

The dataset was examined in Power Query Editor for missing values and logical inconsistencies. No null values were found in key columns such as Spend, Revenue, Clicks, and Conversions. All numerical fields follow logical conditions ($\text{Clicks} \leq \text{Impressions}$, $\text{Conversions} \leq \text{Clicks}$). Hence, the dataset is clean and suitable for analysis.

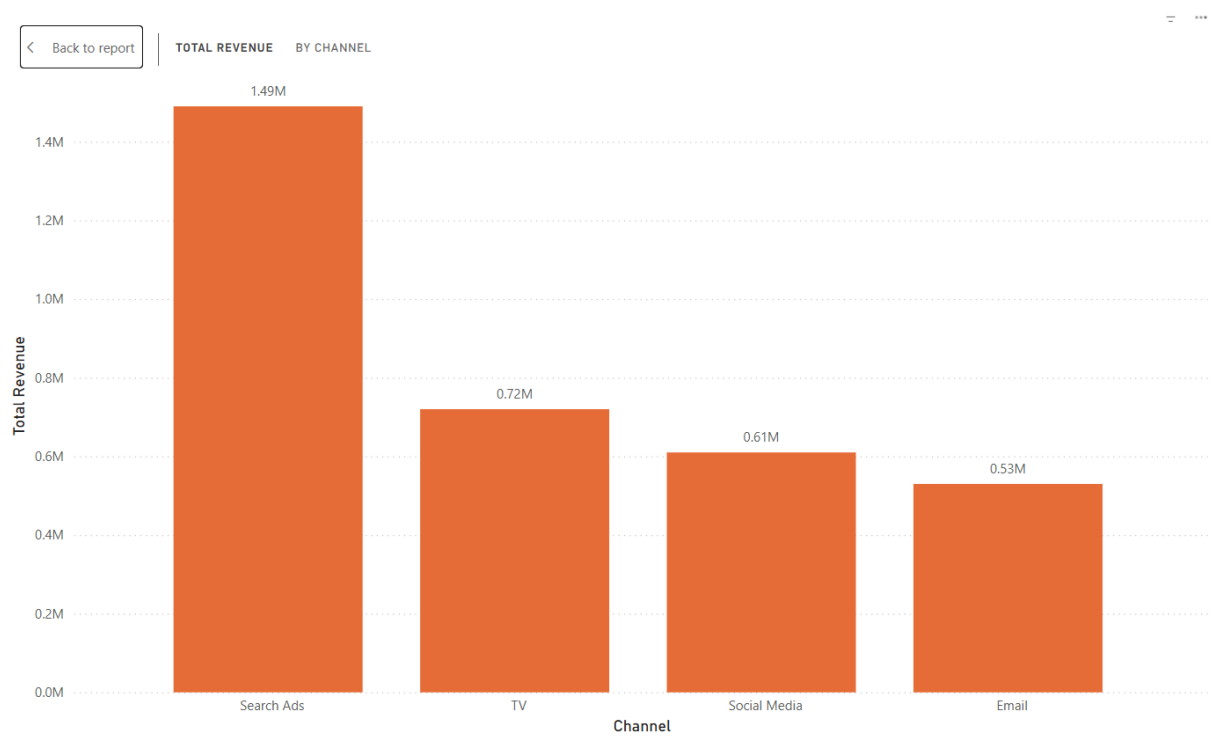
Power Query Editor interface showing a table of marketing campaign data. The table has 12 rows and 6 columns: CampaignID, CampaignName, Channel, StartDate, and EndDate. Each column has a data type dropdown and a status bar showing 100% valid data. The right sidebar shows the 'APPLIED STEPS' list with 'Source', 'Promoted Headers', and 'Changed Type'.

	CampaignID	CampaignName	Channel	StartDate	EndDate
1	C001	New Year Blast	Email	1/1/2024	1/15/2024
2	C002	Winter Sale	Social Media	1/5/2024	1/25/2024
3	C003	Brand Awareness	TV	1/10/2024	2/10/2024
4	C004	Search Boost Q1	Search Ads	1/1/2024	3/31/2024
5	C005	Valentine Offer	Email	2/1/2024	2/14/2024
6	C006	App Install Drive	Social Media	2/10/2024	3/10/2024
7	C007	Summer Promo	TV	3/1/2024	3/31/2024
8	C008	High Intent Search	Search Ads	3/1/2024	3/31/2024
9	C009	Loyalty Email	Email	3/15/2024	3/31/2024
10	C010	Festival Push	Social Media	4/1/2024	4/20/2024
11	C011	Prime Time TV	TV	4/1/2024	4/30/2024
12	C012	Search Retargeting	Search Ads	4/5/2024	4/30/2024

- There will be no error, missing or inconsistent marketing data

2) Which marketing channels generate the highest revenue ?

A revenue analysis by marketing channel shows that Search Ads and TV generate the highest revenue, followed by TV and Email. This indicates that digital channels are more effective in driving revenue compared to traditional marketing channels.



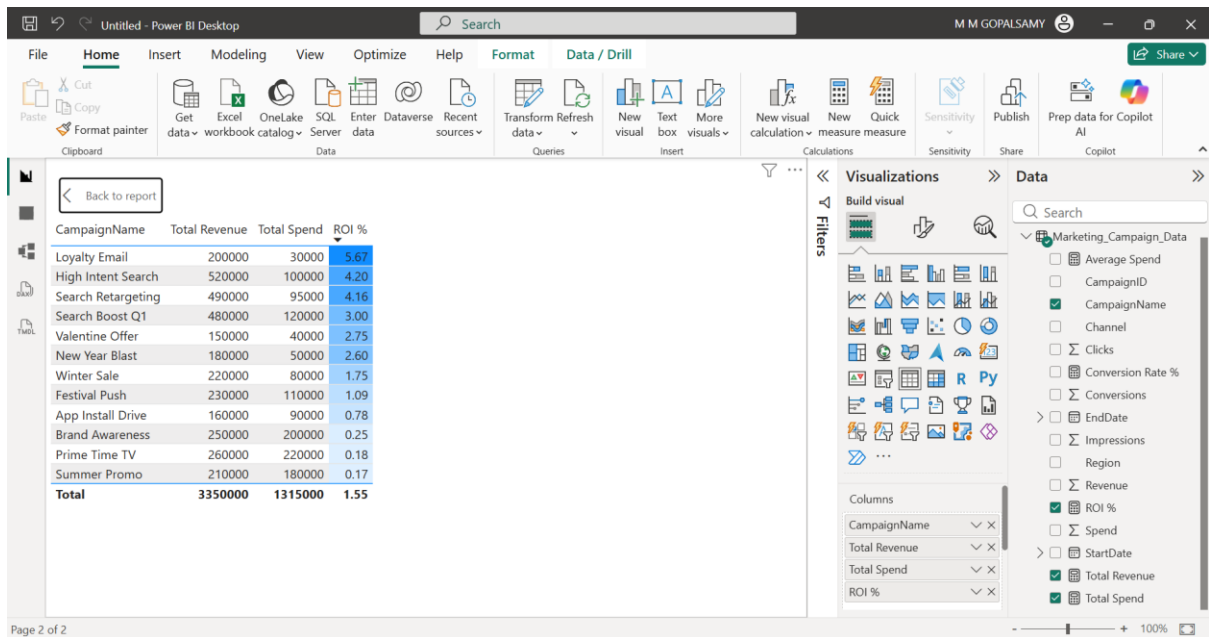
- Search Ads generate the highest revenue

3) Calculate ROI (Return on Investment) for each campaign using DAX ?

ROI was calculated using a DAX measure by comparing total revenue generated against total marketing spend. This measure helps evaluate the profitability of each campaign and identify campaigns delivering positive or negative returns.

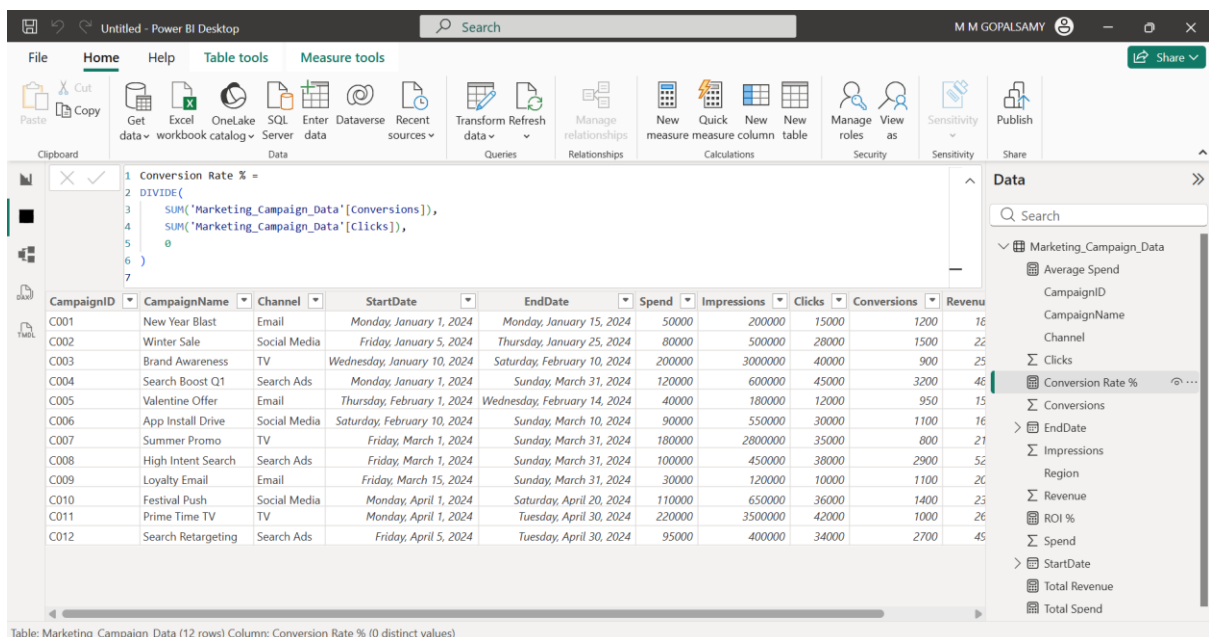
The screenshot shows the Microsoft Power BI Desktop interface. The 'Measure tools' ribbon is active, displaying the DAX formula for 'ROI %' in the formula bar. The formula is:
$$\text{ROI \%} = \frac{[\text{Total Revenue}] - [\text{Total Spend}]}{[\text{Total Spend}]}$$
 The 'Data' pane on the right shows the 'Marketing_Campaign_Data' table with various columns. The main table view displays 12 rows of campaign data, including CampaignID, CampaignName, Channel, StartDate, EndDate, Spend, Impressions, Clicks, Conversions, and Revenue. The 'ROI %' column is highlighted in green, indicating the calculated values for each campaign.

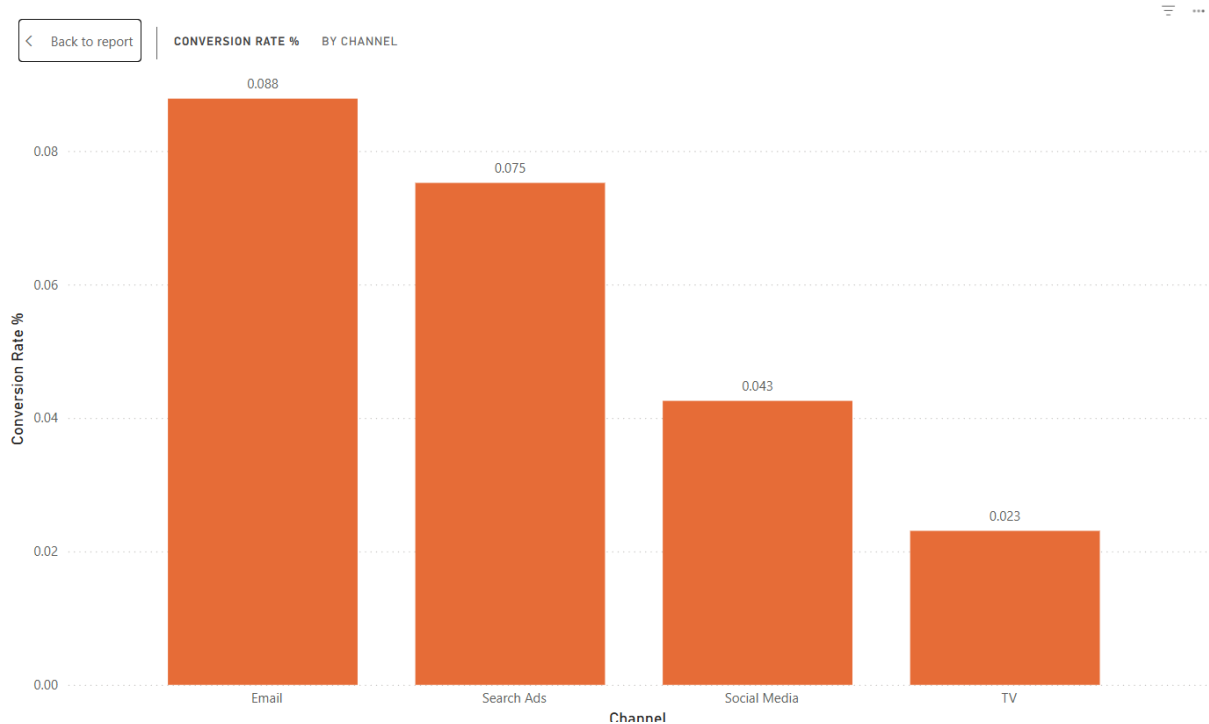
CampaignID	CampaignName	Channel	StartDate	EndDate	Spend	Impressions	Clicks	Conversions	Revenue	ROI %
C001	New Year Blast	Email	Monday, January 1, 2024	Monday, January 15, 2024	50000	200000	15000	1200	16	16
C002	Winter Sale	Social Media	Friday, January 5, 2024	Thursday, January 25, 2024	80000	500000	28000	1500	22	22
C003	Brand Awareness	TV	Wednesday, January 10, 2024	Saturday, February 10, 2024	200000	3000000	40000	900	25	25
C004	Search Boost Q1	Search Ads	Monday, January 1, 2024	Sunday, March 31, 2024	120000	600000	45000	3200	46	46
C005	Valentine Offer	Email	Thursday, February 1, 2024	Wednesday, February 14, 2024	40000	180000	12000	950	15	15
C006	App Install Drive	Social Media	Saturday, February 10, 2024	Sunday, March 10, 2024	90000	550000	30000	1100	16	16
C007	Summer Promo	TV	Friday, March 1, 2024	Sunday, March 31, 2024	180000	2800000	35000	800	21	21
C008	High Intent Search	Search Ads	Friday, March 1, 2024	Sunday, March 31, 2024	100000	450000	38000	2900	52	52
C009	Loyalty Email	Email	Friday, March 15, 2024	Sunday, March 31, 2024	30000	120000	10000	1100	20	20
C010	Festival Push	Social Media	Monday, April 1, 2024	Saturday, April 20, 2024	110000	650000	36000	1400	23	23
C011	Prime Time TV	TV	Monday, April 1, 2024	Tuesday, April 30, 2024	220000	3500000	42000	1000	26	26
C012	Search Retargeting	Search Ads	Friday, April 5, 2024	Tuesday, April 30, 2024	95000	400000	34000	2700	45	45



4)Analyse conversion rates across channels ?

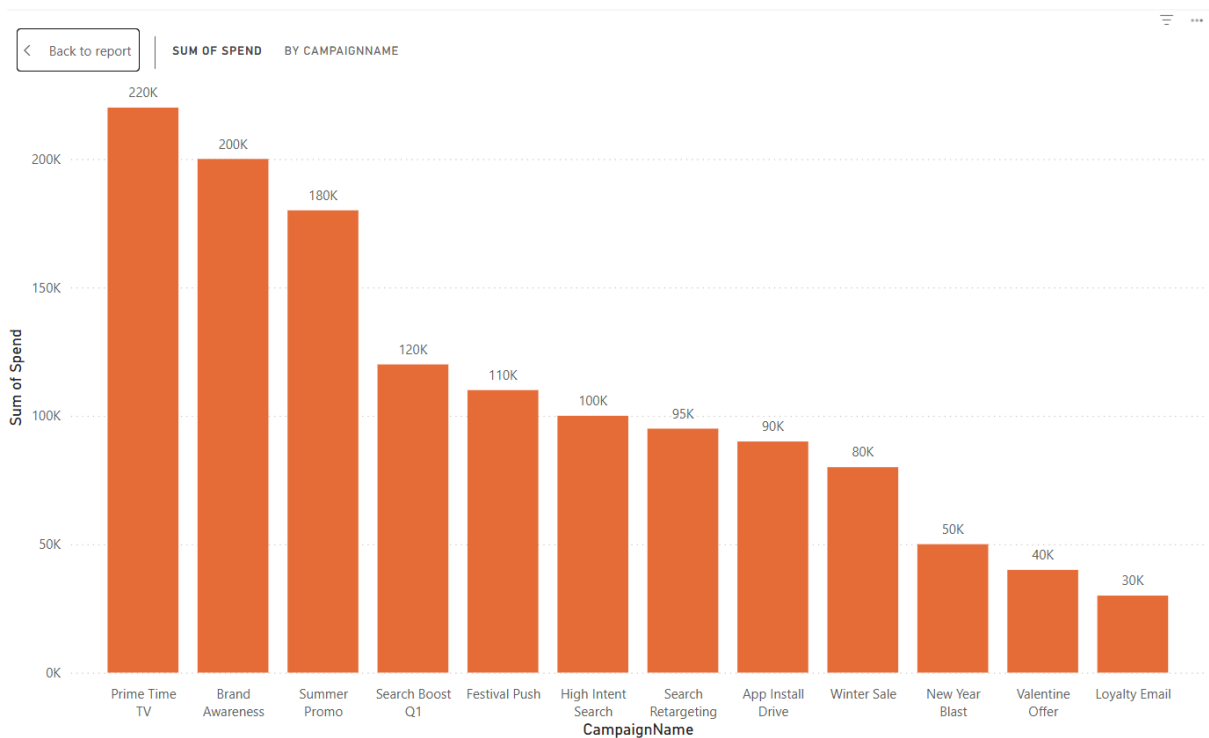
Conversion rate analysis shows that Search Ads and Email Marketing have higher conversion rates, indicating better targeting and customer intent. TV campaigns have the lowest conversion rate, as they are primarily focused on brand awareness rather than direct conversions.





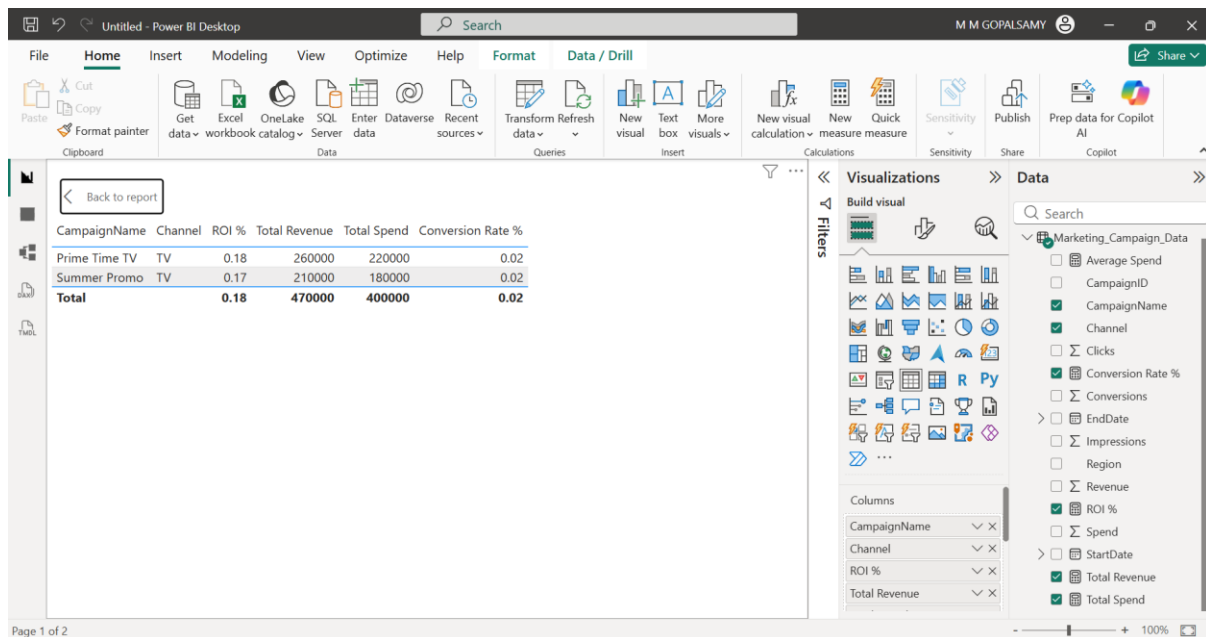
5) Perform univariate analysis on campaign spend ?

Univariate analysis of campaign spend shows that TV campaigns require the highest investment, while Email campaigns operate at a lower cost. This analysis highlights cost-heavy campaigns and helps compare spending patterns across campaigns.



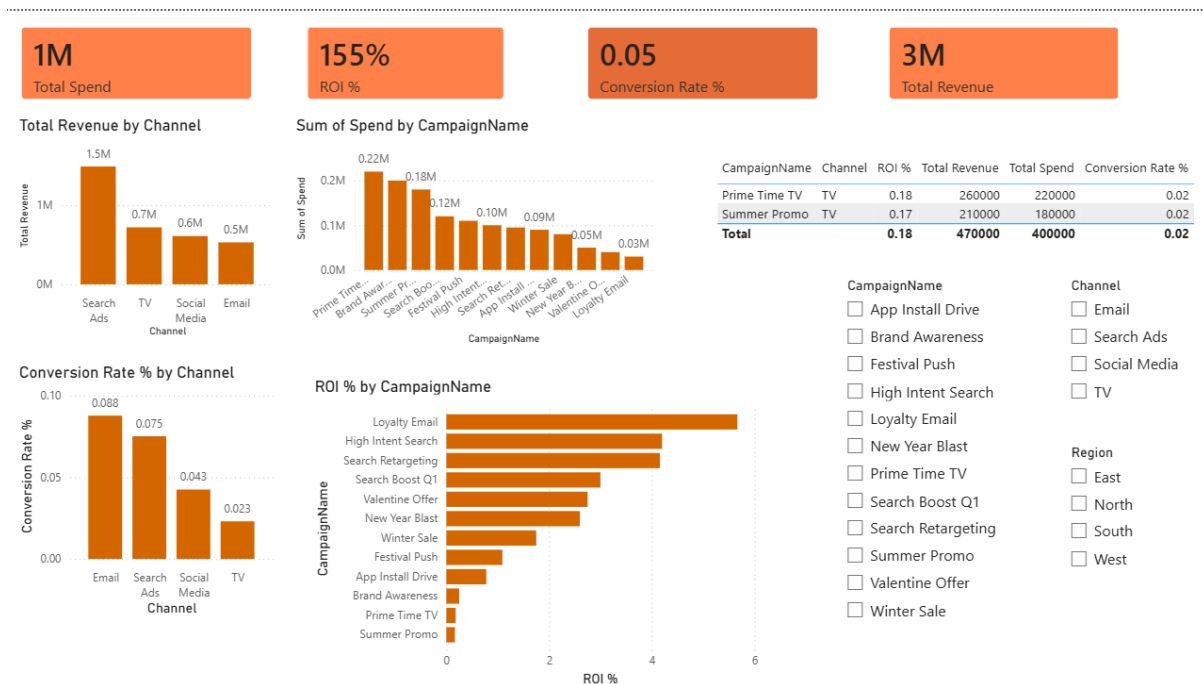
6) Identify underperforming campaigns ?

Campaigns with low ROI and high spend were identified as underperforming. TV-based campaigns and some Social Media campaigns show lower efficiency, indicating the need for optimization or budget reallocation.



7) Design a campaign performance dashboard ?

An interactive Power BI dashboard was designed using KPI cards, bar charts, and tables to display campaign spend, revenue, ROI, and conversion rates. Filters allow users to analyze performance by channel, region, and campaign, enabling data-driven marketing decisions.



8) Which channels should receive increased investment ?

Based on the analysis of ROI, conversion rates, and revenue efficiency, Search Ads and Email Marketing should receive increased investment. These channels demonstrate higher conversion efficiency and positive ROI, indicating effective utilization of marketing spend. Social Media campaigns require further optimization, while TV advertising shows low ROI and should receive reduced investment. Reallocating budget toward high-performing digital channels will improve overall marketing effectiveness.

9) How can insights improve future marketing strategy ?

Insights obtained from campaign performance analysis support data-driven marketing decisions. They enable better budget allocation toward high-performing channels, identification and optimization of underperforming campaigns, and improved targeting and personalization. Continuous monitoring of ROI and conversion rates helps marketers adjust strategies in real time, reduce inefficient spending, and maximize overall marketing effectiveness. This leads to improved return on investment and more successful future marketing campaigns.